

A. Introduction

1. **Title:** Cyber Security — BES Cyber System Categorization
2. **Number:** RRC-CIP-002-1
3. **Purpose:** To identify and categorize BES Cyber Systems (BCS) and their associated BES Cyber Assets (BCA) for the application of cyber security requirements commensurate with the adverse impact that loss, compromise, or misuse of those BCS could have on the reliable operation of the Bulk Electric System (BES). Identification and categorization of BCS that support appropriate protection against compromises that could result in instability, uncontrolled separations, or Cascading within the BES.
4. **Applicability:**
 - 4.1. **Functional Entities:** For the purpose of the requirements contained herein, the following list of functional entities will be collectively referred to as “Responsible Entities.”
 - 4.1.1. **Balancing Authority/Load Balancing Authority**
 - 4.1.2. **Generator Operator**
 - 4.1.3. **Generator Owner**
 - 4.1.4. **Transmission Operator**
 - 4.1.5. **Transmission Owner**
 - 4.1.6. **Distribution Provider:** that owns one or more of the following Facilities, systems, and equipment for the protection or restoration of the BES:
 - 4.1.6.1. Each underfrequency load shedding (UFLS) or undervoltage loadshedding (UVLS) system that:
 - 4.1.6.1.1. is part of a Load shedding program that is subject to one or more requirements in a Railbelt Reliability Standard; and
 - 4.1.6.1.2. performs automatic Load shedding under a common control system owned by the Responsible Entity, without human operator initiation, that the loss, compromise, or misuse could have an adverse impact on the reliable operation of the BES.
 - 4.1.6.2. Remedial Action Scheme (RAS) where the RAS is subject to one or more requirements in a Railbelt Reliability Council Standard.
 - 4.1.6.3. Each Protection System (excluding UFLS and UVLS) that applies to Transmission where the Protection System is subject to one or more requirements in a Railbelt Reliability Council Standard.
 - 4.1.6.4. Each Cranking Path and group of Elements meeting the initial switching requirements from a Blackstart Resource up to and including the first interconnection point of the starting station service of the next generation unit(s) to be started.
 - 4.2. **Facilities:** For the purpose of the requirements contained herein, the following Facilities, systems, and equipment owned by each Responsible Entity in 4.1 above are those to which these requirements are applicable. For requirements in this standard where a specific type of Facilities, system, or equipment or subset of Facilities, systems, and equipment are applicable, these are specified explicitly.
 - 4.2.1. **Distribution Provider:** One or more of the following Facilities, systems and

equipment owned by the Distribution Provider for the protection or restoration of the BES:

4.2.1.1. Each UFLS or UVLS System that:

- 4.2.1.1.1.** is part of a Load shedding program that is subject to one or more requirements in a Railbelt Reliability Council Standard; and
- 4.2.1.1.2.** performs automatic Load shedding under a common control system owned by the Responsible Entity, without human operator initiation, that the loss, compromise, or misuse of could have adverse impact on the reliable operation of the BES.

4.2.1.2. Each RAS where the RAS is subject to one or more requirements in a Railbelt Reliability Council Standard.

4.2.1.3. Each Protection System (excluding UFLS and UVLS) that applies to Transmission where the Protection System is subject to one or more requirements in a Railbelt Reliability Council Standard.

4.2.1.4. Each Cranking Path and group of Elements meeting the initial switching requirements from a Blackstart Resource up to and including the first interconnection point of the starting station service of the next generation unit(s) to be started.

4.2.2. Responsible Entities listed in 4.1 other than Distribution Providers:

All BES Facilities.

4.2.3. Exemptions: The following are exempt from Standard RRC-CIP-002-1:

- 4.2.3.1.** Cyber Systems at Facilities regulated by the Canadian Nuclear Safety Commission.
- 4.2.3.2.** Cyber Assets associated with communication networks and data communication links between discrete Electronic Security Perimeters (ESP).
- 4.2.3.3.** Cyber Systems, associated with communication networks and data communication links, between Cyber Systems, providing confidentiality and integrity of an ESP that extends to one or more geographic locations.
- 4.2.3.4.** The systems, structures, and components that are regulated by the Nuclear Regulatory Commission under a cyber security plan pursuant to 10 C.F.R. Section 73.54.
- 4.2.3.5.** For Distribution Providers, the systems and equipment that are not included in section 4.2.1 above.
- 4.2.3.6.** Responsible Entities that identify that they have no BES Cyber Systems categorized as high impact or medium impact according to the RRC-CIP-002-1 identification and categorization processes.

4.3. “Applicable Systems”: Each table has an “Applicable Systems” column to define the scope of systems to which a specific Requirement Part applies.

5. Effective Dates: Shall become effective on the first day of the first calendar quarter that is twelve (12) months following adoption by the Approving Authority.

B. Requirements and Measures

- R1.** Each Responsible Entity shall implement a process that considers each of the following assets for purposes of parts 1.1 through 1.3: *[Violation Risk Factor: High]*
- i. Control Centers and backup Control Centers;
 - ii. Transmission stations and substations;
 - iii. Generation resources;
 - iv. Systems and facilities critical to system restoration, including Blackstart Resources and Cranking Paths and initial switching requirements;
 - v. RAS that support the reliable operation of the BES; and
 - vi. For Distribution Providers, Protection Systems specified in Applicability section 4.2.1 above.
- 1.1.** Identify each of the high impact BCS according to Attachment 1, Section 1, if any, at each asset;
- 1.2.** Identify each of the medium impact BCS according to Attachment 1, Section 2, if any, at each asset; and
- 1.3.** Identify each asset that contains a low impact BCS according to Attachment 1, Section 3, if any (a discrete list of low impact BCS is not required).
- M1.** Acceptable evidence includes, but is not limited to, dated electronic or physical lists required by Requirement R1, and Parts 1.1 and 1.2.
- R2.** The Responsible Entity shall: *[Violation Risk Factor: Lower]*
- 2.1** Review the identifications in Requirement R1 and its parts (and update them if there are changes identified) at least once every 15 calendar months, even if it has no identified items in Requirement R1, and
 - 2.2** Have its CIP Senior Manager or delegate approve the identifications required by Requirement R1 at least once every 15 calendar months, even if it has no identified items in Requirement R1.
- M2.** Acceptable evidence includes, but is not limited to, electronic or physical dated records to demonstrate that the Responsible Entity has reviewed and updated, where necessary, the identifications required in Requirement R1 and its parts, and has had its CIP Senior Manager or delegate approve the identifications required in Requirement R1 and its parts at least once every 15 calendar months, even if it has none identified in Requirement R1 and its parts, as required by Requirement R2.

C. Compliance

1. Compliance Monitoring Process

1.1 Compliance Enforcement Authority:

The RRC is the entity with the authority to monitor and enforce compliance with mandatory and enforceable Railbelt Reliability Council Standards.

1.2 Evidence Retention:

The following evidence retention periods identify the period of time an entity is required to retain specific evidence to demonstrate compliance. For instances where the evidence retention period specified below is shorter than the time since the last audit, the RRC may ask an entity to provide other evidence to show that it was compliant for the full time period since the last audit.

The Responsible Entity shall keep data or evidence to show compliance as identified below unless directed by the RRC to retain specific evidence for a longer period of time as part of an investigation:

- Each Responsible Entity shall retain evidence of each requirement in this standard for three calendar years.
- If a Responsible Entity is found non-compliant, it shall keep information related to the non-compliance until mitigation is complete and approved or the time specified above, whichever is longer.
- The RRC shall keep the last audit records and all requested and submitted subsequent audit records.

1.3 Compliance Monitoring and Enforcement Program:

Compliance Monitoring and Enforcement Program refers to the identification of the processes that will be used to evaluate data or information for the purpose of assessing performance or outcomes with the associated Railbelt Reliability Council Standards.

Table of Compliance Elements

R#	Violation Risk Factor (VRF)	Violation Severity Levels (RRC-CIP-002-1)			
		Lower VSL	Moderate VSL	High VSL	Severe VSL
R1	High	For Responsible Entities, five percent or fewer BES assets have not been considered according to Requirement R1; OR For Responsible Entities, five percent or fewer of identified BES assets have not been categorized or have been incorrectly categorized at a lower category; OR For Responsible Entities, five percent or fewer high or medium BCS have not been identified; OR For Responsible Entities, five percent or fewer of identified high or medium	For Responsible Entities, more than five percent but less than or equal to 10 percent of BES assets have not been considered, according to Requirement R1; OR For Responsible Entities, more than five percent but less than or equal to 10 percent of identified BES assets have not been categorized or have been incorrectly categorized at a lower category; OR For Responsible Entities, more than five percent but less than or equal to 10 percent of high or medium BCS have not been identified; OR	For Responsible Entities, more than 10 percent but less than or equal to 15 percent of BES assets have not been considered, according to Requirement R1; OR For Responsible Entities, more than 10 percent but less than or equal to 15 percent of identified BES assets have not been categorized or have been incorrectly categorized at a lower category; OR For Responsible Entities, more than 10 percent but less than or equal to 15 percent of high or medium BCS have not been identified; OR	For Responsible Entities, more than 15 percent of BES assets have not been considered, according to Requirement R1; OR For Responsible Entities, more than 15 percent but less than or equal to 20 percent of identified BES assets have not been categorized or have been incorrectly categorized at a lower category; OR For Responsible Entities, more than 15 percent but less than or equal to 20 percent of high or medium BCS have not been identified; OR

R#	Violation Risk Factor (VRF)	Violation Severity Levels (RRC-CIP-002-1)			
		Lower VSL	Moderate VSL	High VSL	Severe VSL
		BCS have not been categorized or have been incorrectly categorized at a lower category.	For Responsible Entities, more than five percent but less than or equal to 10 percent identified high or medium BCS have not been categorized or have been incorrectly categorized at a lower category.	For Responsible Entities, more than 10 percent but less than or equal to 15 percent of identified high or medium BCS have not been categorized or have been incorrectly categorized at a lower category.	For Responsible Entities, more than 15 percent but less than or equal to 20 percent of identified high or medium BCS have not been categorized or have been incorrectly categorized at a lower category.

R#	Violation Risk Factor (VRF)	Violation Severity Levels (RRC-CIP-002-1)			
		Lower VSL	Moderate VSL	High VSL	Severe VSL
R2	Lower	The Responsible Entity did not complete its review and update for the identification required for Requirement R1 within 15 calendar months but less than or equal to 18 calendar months of the previous review. (R 2.1) OR The Responsible Entity did not complete its approval of the identifications required by Requirement R1 by the CIP Senior Manager or delegate according to Requirement R2 within 15 calendar months but less than or equal to 18 calendar months of the previous approval. (R 2.2)	The Responsible Entity did not complete its review and update for the identification required for Requirement R1 within 18 calendar months but less than or equal to 21 calendar months of the previous review. (R 2.1) OR The Responsible Entity failed to complete its approval of the identifications required by Requirement R1 by the CIP Senior Manager or delegate according to Requirement R2 within 18 calendar months but less than or equal to 21 calendar months of the previous approval. (R 2.2)	The Responsible Entity did not complete its review and update for the identification required for Requirement R1 within 21 calendar months but less than or equal to 24 calendar months of the previous review. (R 2.1) OR The Responsible Entity failed to complete its approval of the identifications required by Requirement R1 by the CIP Senior Manager or delegate according to Requirement R2 within 21 calendar months but less than or equal to 24 calendar months of the previous approval. (R 2.2)	The Responsible Entity did not complete its review and update for the identification required for Requirement R1 within 24 calendar months of the previous review. (R 2.1) OR The Responsible Entity failed to complete its approval of the identifications required by Requirement R1 by the CIP Senior Manager or delegate according to Requirement R2 within 24 calendar months of the previous approval. (R 2.2)

D. Regional Variances

None.

E. Interpretations

None.

F. Associated Documents

None

Version History

Version	Date	Action	Change Tracking

RRC-CIP-002-1 - Attachment 1**Impact Rating Criteria**

The criteria defined in Attachment 1 are criteria characterizing the level of impact and are referenced by requirements.

1. High Impact Rating (H)

Each BCS used by and located at any of the following:

- 1.1.** Each Control Center or backup Control Center used to perform the functional obligations of the Balancing Authority.
- 1.2.** Each Control Center or backup Control Center used to perform the functional obligations of the Balancing Authority/Load Balancing Authority: 1) for generation equal to or greater than an aggregate of 120 MVA in a single Interconnection, or 2) for one or more of the assets that meet criterion 2.3 or 2.8.
- 1.3.** Each Control Center or backup Control Center used to perform the functional obligations of the Transmission Operator for one or more of the assets that meet criterion 2.2, 2.4, 2.5, 2.7.
- 1.4** Each Control Center or backup Control Center used to perform the functional obligations of the Generator Operator for one or more of the assets that meet criterion 2.1, 2.3, 2.8.

2. Medium Impact Rating (M)

Each BCS, not included in Section 1 above, associated with any of the following:

- 2.1.** Commissioned generation, by each group of generating units at a single plant location, that is equal to or larger than the largest single load impacting contingency. The largest single load impacting contingency is typically determined as part of the RRC- BAL-002 O&P Standard. For each group of generating units, the only BCS that meet this criterion are each discrete shared BCS that could, within 15 minutes, adversely impact the reliable operation of any combination of units that in aggregate equal or exceed the single largest contingency.
- 2.2.** Each BES reactive resource, or group of resources at a single location (excluding generation Facilities) with an aggregate maximum Reactive Power nameplate rating of 40 MVAR or greater (excluding those at generation Facilities). The only BCS that meet this criterion are ~~those~~ each discrete shared BCS that could, within 15 minutes, adversely impact the reliable operation of any combination of resources that in aggregate equal or exceed 40 MVAR.
- 2.3.** Each generation Facility that its Planning Coordinator designates, and informs the Generator Owner or Generator Operator, as necessary to avoid an Adverse Reliability Impact in the planning horizon of more than one year.
- 2.4.** Transmission Facilities operated at 300 kV or higher. For this purpose of this criterion, the collector bus for a generation plant is not considered a Transmission Facility but is part of the generation interconnection Facility.
- 2.5.** Transmission Facilities that are operating between 100 kV and 299 kV at a single station or substation, where the station or substation is connected at 100 kV or higher voltages to three or more other Transmission stations or substations and has an "aggregate weighted value" exceeding 2700 according to the table below. The "aggregate weighted value" for a single station or substation is determined by summing the "weight value per line" shown in the table below for each incoming and each outgoing BES Transmission Line that is connected to another Transmission station or substation. For the purpose of this criterion, the collector bus for a generation plant is not considered a Transmission Facility but is part of the generation

interconnection Facility.

Voltage Value of a Line	Weight Value per Line
less than 100 kV (not applicable)	(not applicable)
100 kV to 199 kV	700
200 kV to 299 kV	1300

- 2.6.** Generation at a single plant location or Transmission Facilities at a single station or substation location that are identified by the Planning Authority as critical to the derivation of Interconnection Reliability Operating Limits (IROLs) and their associated contingencies.
- 2.7.** Transmission Facilities, including generation interconnection Facilities, providing the generation interconnection required to connect generator output to the Transmission Systems that, if destroyed, degraded, misused, or otherwise rendered unavailable, would result in the loss of the generation Facilities identified by any Generator Owner as a result of its application of Attachment 1, criterion 2.1 or 2.3.
- 2.8.** Each Remedial Action Scheme (RAS), or automated switching System that operates BES Elements, that, if destroyed, degraded, misused or otherwise rendered unavailable, would cause one or more Interconnection Reliability Operating Limits (IROLs) violations for failure to operate as designed or cause a reduction in one or more IROLs if destroyed, degraded, misused, or otherwise rendered unavailable.
- 2.9.** Each system or group of Elements that performs automatic Load shedding under a common control system, without human operator initiation, of 15 MVA or more implementing undervoltage load shedding (UVLS) or underfrequency load shedding (UFLS) under a load shedding program that is subject to one or more requirements in a Railbelt Reliability Council Standard.
- 2.10.** Each Control Center or backup Control Center, not already included in High Impact Rating (H) above, used to perform the functional obligations of the Generator Operator for an aggregate highest rated net Real Power capability of the preceding 12 calendar months equal to the single largest generation contingency in a single Interconnection.
- 2.11.** Each Control Center or backup Control Center used to perform the functional obligations of the Transmission Operator not included in High Impact Rating (H), above.
- 2.12.** Each Control Center or backup Control Center, not already included in High Impact Rating (H) above, used to perform the functional obligations of the Balancing Authority for generation equal to or greater than an aggregate of the single largest generation contingency in a single Interconnection.

3. Low Impact Rating (L)

BCS not included in Sections 1 or 2 above that are associated with any of the following assets and that meet the applicability qualifications in Section 4 - Applicability, part 4.2 – Facilities, of this standard:

- 3.1.** Control Centers and backup Control Centers.
- 3.2.** Transmission stations and substations.
- 3.3.** Generation resources.
- 3.4.** Systems and facilities critical to system restoration, including Blackstart Resources and Cranking Paths and initial switching requirements.
- 3.5.** RAS that support the reliable operation of the BES.

- 3.6.** For Distribution Providers, Protection Systems specified in Applicability section 4.2.1 above.

GLOSSARY

Term	Definition
Balancing Authority	The responsible entity that integrates resource plans ahead of time, maintains Demand and resource balance within a Balancing Authority Area, and supports Interconnection frequency in real time.
BES Cyber Asset (BCA)	A Cyber Asset that if rendered unavailable, degraded, or misused would, within 15 minutes of its required operation, misoperation, or non operation, adversely impact one or more Facilities, systems, or equipment, which, if destroyed, degraded, or otherwise rendered unavailable when needed, would affect the reliable operation of the Bulk Electric System. Redundancy of affected Facilities, systems, and equipment shall not be considered when determining adverse impact. Each BES Cyber Asset is included in one or more BES Cyber Systems.
BES Cyber System (BCS)	One or more BES Cyber Assets logically grouped by a responsible entity to perform one or more reliability tasks for a functional entity.
BES Cyber System Information	Information about the BES Cyber System that could be used to gain unauthorized access or pose a security threat to the BES Cyber System. BES Cyber System Information does not include individual pieces of information that by themselves do not pose a threat or could not be used to allow unauthorized access to BES Cyber Systems, such as, but not limited to, device names, individual IP addresses without context, ESP names, or policy statements. Examples of BES Cyber System Information may include, but are not limited to, security procedures or security information about BES Cyber Systems, Physical Access Control Systems, and Electronic Access Control or Monitoring Systems that is not publicly available and could be used to allow unauthorized access or unauthorized distribution; collections of network addresses; and network topology of the BES Cyber System.
Blackstart Resource	A generating unit(s) and its associated set of equipment which has the ability to be started without support from the System or is designed to remain energized without connection to the remainder of the System, with the ability to energize a bus, meeting the Transmission Operator's restoration plan needs for Real and Reactive Power capability, frequency and voltage control, and that has been included in the Transmission Operator's restoration plan.
Bulk Electric System (BES)	Unless modified by the lists shown below, all Transmission Elements operated at 100 kV or higher and Real Power and Reactive Power resources connected at 100 kV or higher. This does not include facilities used in the local distribution of electric energy. Inclusions: • I1 - Transformers with the primary terminal and at least one secondary terminal operated at 100 kV or higher unless excluded by application of Exclusion E1 or E3. • I2 – Generating resource(s) including the generator terminals through the high-side of the step-up transformer(s) connected at a voltage of 100 kV or above with: a) Gross individual nameplate rating greater than 20 MVA. Or, b) Gross plant/facility aggregate nameplate rating greater than 75 MVA. • I3 - Blackstart Resources identified in the Transmission

	Operator's restoration plan. • I4 - Dispersed power producing resources that aggregate to a total capacity greater than 75 MVA (gross nameplate rating), and that are connected through a system designed primarily for delivering such capacity to a common point of connection at a voltage of 100 kV or above. Thus, the facilities designated as BES are: a) The individual resources, and The system designed primarily for delivering capacity from the point where those resources aggregate to greater than 75 MVA to a common point of connection at a voltage of 100 kV or above. • I5 –Static or dynamic devices (excluding generators) dedicated to supplying or absorbing Reactive Power that are connected at 100 kV or higher, or through a dedicated transformer with a high-side voltage of 100 kV or higher, or through a transformer that is designated in Inclusion I1 unless excluded by application of Exclusion E4 Exclusions: • E1 - Radial systems: A group of contiguous transmission Elements that emanates from a single point of connection of 100 kV or higher and: a) Only serves Load. Or, b) Only includes generation resources, not identified in Inclusions I2, I3, or I4, with an aggregate capacity less than or equal to 75 MVA (gross nameplate rating). Or, c) Where the radial system serves Load and includes generation resources, not identified in Inclusions I2, I3 or I4, with an aggregate capacity of non-retail generation less than or equal to 75 MVA (gross nameplate rating). Note 1 – A normally open switching device between radial systems, as depicted on prints or one-line diagrams for example, does not affect this exclusion. Note 2 – The presence of a contiguous loop, operated at a voltage level of 50 kV or less, between configurations being considered as radial systems, does not affect this exclusion. • E2 - A generating unit or multiple generating units on the customer's side of the retail meter that serve all or part of the retail Load with electric energy if: (i) the net capacity provided to the BES does not exceed 75 MVA, and (ii) standby, back-up, and maintenance power services are provided to the generating unit or multiple generating units or to the retail Load by a Balancing Authority, or provided pursuant to a binding obligation with a Generator Owner or Generator Operator, or under terms approved by the applicable regulatory authority. • E3 - Local networks (LN): A group of contiguous transmission Elements operated at less than 300 kV that distribute power to Load rather than transfer bulk power across the interconnected system. LN's emanate from multiple points of connection at 100 kV or higher to improve the level of service to retail customers and not to accommodate bulk power transfer across the interconnected system. The LN is characterized by all of the following: a) Limits on connected generation: The LN and its underlying Elements do not include generation resources identified in Inclusions I2, I3, or I4 and do not have an aggregate capacity of non-retail generation greater than 75 MVA (gross nameplate rating); b) Real Power flows only into the LN and the LN does not transfer energy originating outside the LN for delivery through the LN; and c) Not part of a Flowgate or transfer path: The LN does not contain any
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Glossary

	part of a permanent Flowgate in the Eastern Interconnection, a major transfer path within the Western Interconnection, or a comparable monitored Facility in the ERCOT or Quebec Interconnections, and is not a monitored Facility included in an Interconnection Reliability Operating Limit (IROL). • E4 – Reactive Power devices installed for the sole benefit of a retail customer(s). Note - Elements may be included or excluded on a case-by-case basis through b) the Rules of Procedure exception process.
CIP Exceptional Circumstance	A situation that involves or threatens to involve one or more of the following, or similar, conditions that impact safety or BES reliability: a risk of injury or death; a natural disaster; civil unrest; an imminent or existing hardware, software, or equipment failure; a Cyber Security Incident requiring emergency assistance; a response by emergency services; the enactment of a mutual assistance agreement; or an impediment of large scale workforce availability.
CIP Senior Manager	A single official with the authority to bind the registered entity to adhere to the requirements within the CIP Standards, CIP-002 through CIP-015.
Control Center	One or more facilities hosting operating personnel that monitor and control the Bulk Electric System (BES) in real-time to perform the reliability tasks, including their associated data centers, of: 1) a Reliability Coordinator, 2) a Balancing Authority, 3) a Transmission Operator for transmission Facilities at two or more locations, or 4) a Generator Operator for generation Facilities at two or more locations.
Cranking Path	A portion of the electric system that can be isolated and then energized to deliver electric power from a generation source to enable the startup of one or more other generating units.
Cyber Assets	Programmable electronic devices, including the hardware, software, and data in those devices.
Distribution Provider	Provides and operates the “wires” between the transmission system and the end-use customer. For those end-use customers who are served at transmission voltages, the Transmission Owner also serves as the Distribution Provider. Thus, the Distribution Provider is not defined by a specific voltage, but rather as performing the distribution function at any voltage
Electronic Access Control or Monitoring Systems	Cyber Assets that perform electronic access control or electronic access monitoring of the Electronic Security Perimeter(s) or BES Cyber Systems. This includes Intermediate Systems.
Electronic Access Point	A Cyber Asset interface on an Electronic Security Perimeter that allows routable communication between Cyber Assets outside an Electronic Security Perimeter and Cyber Assets inside an Electronic Security Perimeter.
Electronic Security Perimeter	The logical border surrounding a network to which BES Cyber Systems are connected using a routable protocol.
Element	Any electrical device with terminals that may be connected to other electrical devices such as a generator, transformer, circuit breaker, bus section, or transmission line. An Element may be comprised of one or more components.